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/ [\[Exam\] Connections between control and inference](#)

Begonnen am Thursday, 25. May 2023, 17:06

Status Beendet

Beendet am Thursday, 25. May 2023, 17:16

Verbrauchte Zeit 9 Minuten 17 Sekunden

Punkte 1,57/7,00

Bewertung 2,24 von 10,00 (22,38%)

Frage 1

Vollständig

Erreichte Punkte 0,00 von 1,00

What are the motivations for a probabilistic framework in RL?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Probabilistics models can explain better optimal behaviours
- ☒ b. Probabilistics models can explain better stochastic behaviours
- ☐ c. Traditional RL framework does not explain optimality
- ☐ d. Traditional RL framework does not explain sub optimality

Frage 2

Vollständig

Erreichte Punkte 0,00 von 1,00

What is a probabilistic graphical model?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. It's a model that uses stochastic processes to solve Partial Observable MDPs (POMDPs)
- ☐ b. It's a probabilistic model for which a graph expresses the conditional dependence structure between random variables
- ☐ c. It's a probabilistic model for which a graph expresses the sub-optimal behaviours
- ☐ d. It's a model that uses stochastic processes to solve MDPs

Frage 3

Vollständig

Erreichte Punkte 1,00 von 1,00

What is the connection between optimal control (OC) and RL, and probabilistic inference (PI), as seen in class?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. OC and RL are fundamentally different to PI, but the underlying dynamical system can be stochastic.
- ☒ b. OC and RL are fundamentally different to PI, but the underlying dynamical system can be described by the same model.
- ☐ c. OC and RL are fundamentally different to PI, but all learn a policy

Frage 4

Vollständig

Erreichte Punkte 0,00 von 1,00

What of the following statements are correct when referring to the "optimality variable"

Wählen Sie eine oder mehrere Antworten:

- ☐ a. We use them to infer other random variables on the condition (of the optimal variables) to be true.
- ☐ b. They are observations in a Hidden Markov Model style framework
- ☐ c. They measure how far from optimal a trajectory is
- ☒ d. They measure how optimal a trajectory is

Frage 5

Vollständig

Erreichte Punkte 0,50 von 1,00

According to this formula,

$$p(\tau \mid \mathcal{O}_{1:T}) \propto \left[p(s_0) \prod_{t=0}^T p(s_{t+1} \mid s_t, a_t) \right] \exp\left(\sum_{t=0}^T r(s_t, a_t) \right) \\ \propto p(\tau) \exp\left(\sum_t r(s_t, a_t) \right)$$

The probability of a trajectory is given by:

$$p(\tau) = p(s_0) \prod_{t=0}^T p(s_{t+1} \mid s_t, a_t)$$

Why the policy does not appear in the formula? Is it wrong?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Since the trajectory is not conditioned on being optimal, one can assume that the policy is random. The formula is fine within this framework.
- ☐ b. Both formulas are wrong. The policy term is missing in both.
- ☐ c. Since the trajectory depends both on the model and the policy, both equations are wrong.
- ☐ d. Nothing is wrong with the formula. It is assuming that the policy is random.

Frage 6

Vollständig

Erreichte Punkte 0,00 von 1,00

After recovering the policy in the Probabilistic Inference framework (as seen in class), we have the following:

$$\pi(a_t | s_t) = \exp(A(s_t, a_t))$$

How is it possible that the policy becomes the exponential of the advantage function??

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Because the policy is the ratio between backward messages $\frac{\beta(s_t, a_t)}{\beta(s_t)}$. After some rearrangements we get something like $(f(s, a) - f(s))$. Since both $f(s, a)$ and $f(s)$ are in log space, that's exactly the exponential of the advantage function.
- ☐ b. Is not the same advantage function that we know from classical RL. The policy is the ratio between backward messages $\frac{\beta(s_t)}{\beta(s_t, a_t)}$. After some rearrangements we get something like $\log(f(s)) - \log(f(s, a))$. If we get the exponential of this expression we get something which looks like the advantage function we know.
- ☒ c. Because the policy is the ratio between backward messages $\frac{\beta(s_t)}{\beta(s_t, a_t)}$. After some rearrangements we get something like $(f(s) - f(s, a))$. Since both $f(s)$ and $f(s, a)$ are in log space, that's exactly the exponential of the advantage function.
- ☐ d. Is not the same advantage function that we know from classical RL. The policy is the ratio between backward messages $\frac{\beta(s_t, a_t)}{\beta(s_t)}$. After some rearrangements we get something like $\log(f(s, a)) - \log(f(s))$. If we get the exponential of this expression we get something which looks like the advantage function we know.

Frage 7

Vollständig

Erreichte Punkte 0,07 von 1,00

What are the benefits of soft-optimality?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Good for deterministic behaviours
- ☒ b. It reduces overestimation
- ☒ c. Improve exploration
- ☐ d. Better robustness
- ☐ e. Good model for human behaviour
- ☐ f. Easier to specialize (finetune) policies for specific task
- ☒ g. Prevent entropy collapse
- ☐ h. Better generalization capabilities

◀ Final questions

Direkt zu:

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